Lead 16.3: the vertical security margin

Round 4, 2026-10-11. Code `examples/r4_16_3_vertical_margin.rs` (native Rust), base commit `440c0e163`, branch `research/isogeny-fibers-round4`. Raw records: `rational.json`, `generic.json`, `elkies.json`; fits and derived table: `analysis.json` / `analysis.txt`. Full write-up: `RESULTS.md`. Toy, synthetic instances only; no sub-rho or deployed-curve claim.

Question

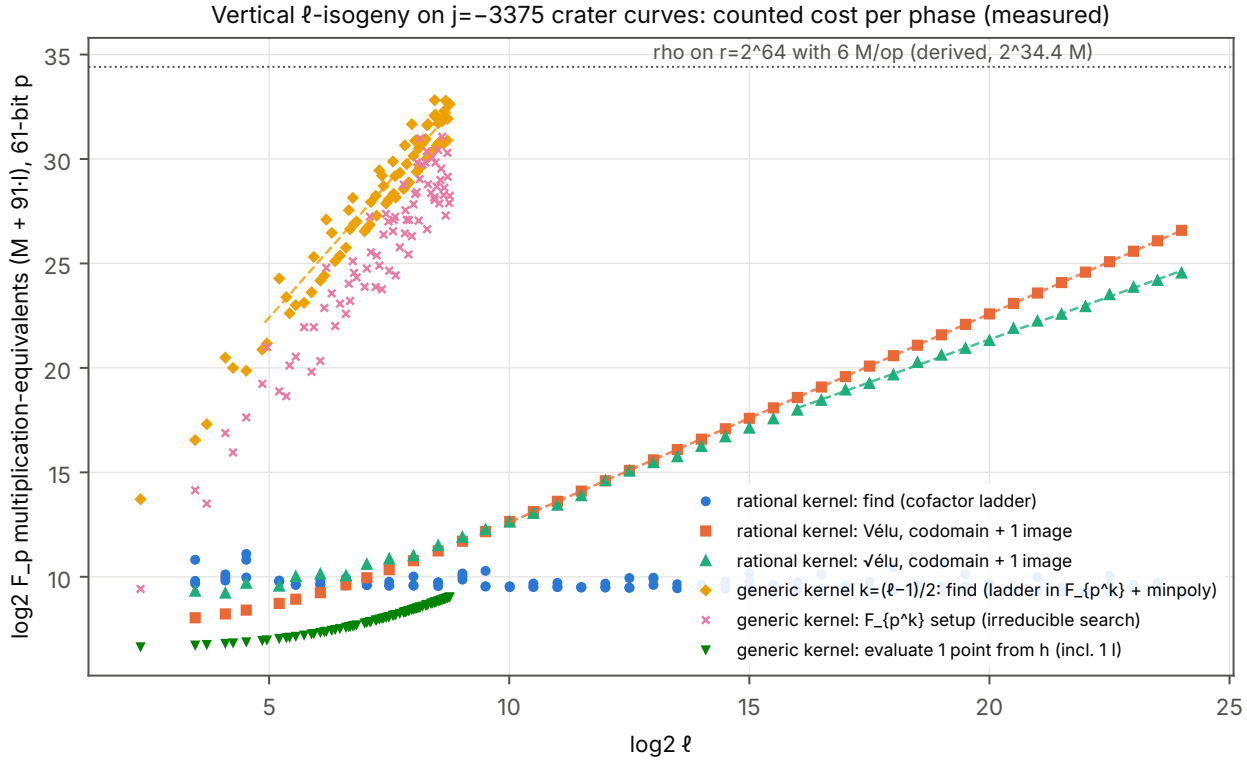
Above which prime ℓ in a conductor gap does a vertical transfer (find the kernel + evaluate the isogeny) cost more than rho on the target, $\sqrt{\pi r/4}$ group operations? Prop. 3.3 predicts finding $\tilde{O}(\ell^2)$, evaluating $\tilde{O}\left(\sqrt{\ell}\right)$ (√élu) or $O(\ell)$ (Vélu).

Method (measured)

Instances: the $j = -3375$ curve $y^2 = x^3 - 35x - 98$ (CM by the maximal order of $\mathbb{Q}\left(\sqrt{-7}\right)$) over primes $p = (t^2 + 7v^2)/4 \in [2^{60}, 2^{61})$ with $\ell \mid v$ and $t \equiv 2\lambda \pmod{\ell}$: the crater of an ℓ-volcano, Frobenius scalar λ on $E[\ell]$, so the $\ell - (-7/\ell)$ non-horizontal lines are descending (vertical) isogenies. A codomain with $j \neq -3375$ certifies a vertical edge.

- **Rational kernel** ($\lambda = 1$, $E[\ell] \subset E(\mathbb{F}_p)$, $\ell^2 \parallel \#E$): find = cofactor ladder; evaluate by Vélu and by √élu (BDLS 2020), $\ell = 11 \ldots 2^{24}$, 3 instances each.
- **Generic kernel** (λ primitive, kernel x-coordinates generate $\mathbb{F}_{p^k}$, $k = (\ell - 1)/2$): find = cofactor ladder on the twist over $\mathbb{F}_{p^k}$ + Berlekamp–Massey minimal polynomial; evaluate from the kernel polynomial. $\ell = 5 \ldots 433$ (420 s cap).
- **Library Elkies route** (`isogeny_walk`: Φ_ℓ mod p, roots, Elkies kernel): contended wall time, $\ell \leq 79$ (300 s cap).

Unit: M = $\mathbb{F}_p$ multiplications and squarings (inside polynomial and $\mathbb{F}_{p^k}$ arithmetic too; Karatsuba, threshold 24); I = inversions, Meq $= M + 91I$. Checks (all passed): √élu = Vélu on codomain and images (126 rows); images have order dividing $\#E$ on the codomain; library `verify_kernel` + equal j (24 rational rows, 44 generic rows); Elkies succeeded on every non-crater root of $\Phi_\ell(-3375, Y)$.



Dashed: OLS fits from analysis.json. Rational: 3 instances per ℓ (Vélu/√élu counts are identical across instances). Generic: 1 instance per ℓ.

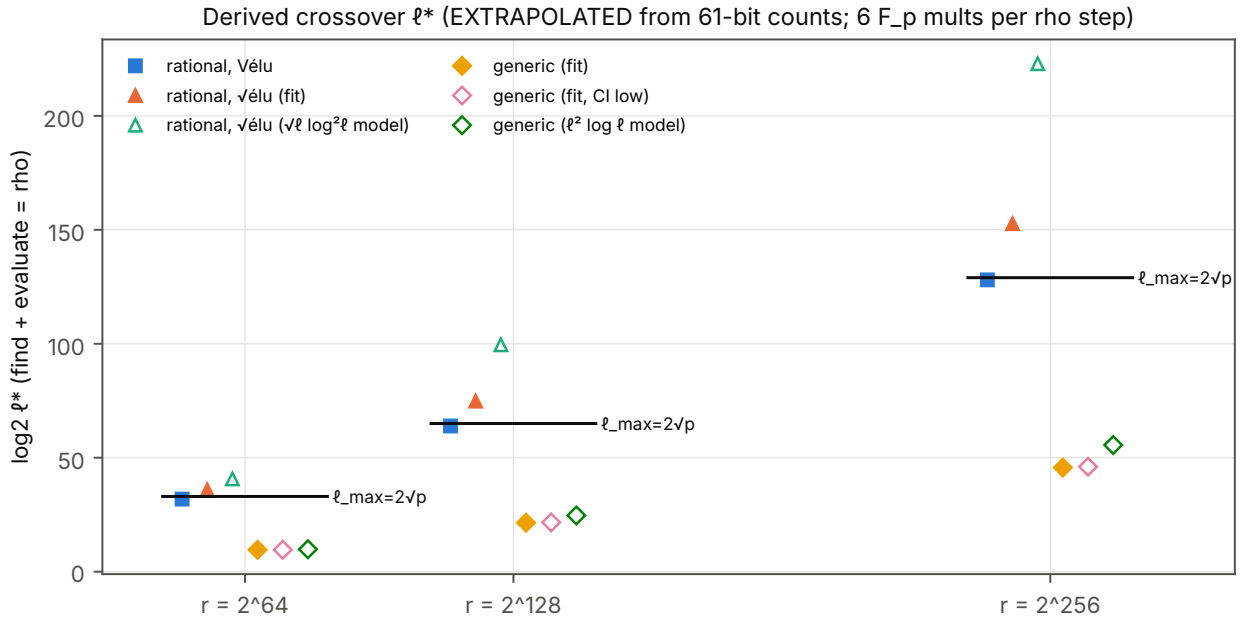
Figure 1: Counted cost per phase against ℓ (measured, 61-bit p). Dashed: OLS fits. Dotted: rho on $r = 2^{64}$ at 6 M per step (derived).

Fitted exponents (measured; 95 % t-intervals measure power-law misfit, counts are deterministic)

phase	range	exponent [95 % CI]	$\log_2$ const
rational find	$11 \ldots 2^{24}$	−0.025 [−0.036, −0.015] (flat, ≤ 1,663 Meq)	10.01
Vélu, codomain + 1 image	$\geq 2^{10}$	0.998 [0.997, 0.999]	2.62
√élu, codomain + 1 image	$\geq 2^{10}$	0.856 [0.845, 0.868]	4.22
√élu, codomain + 1 image	$\geq 2^{16}$	0.822 [0.805, 0.838]	4.93
generic find, expected	$31 \ldots 433$	2.605 [2.578, 2.631]	9.38

generic find, expected	≥ 150	2.552 [2.477, 2.628]	9.81
generic $\mathbb{F}_{p^k}$ setup	31 … 433	2.831 [2.547, 3.115]	4.94
generic eval / point (M)	31 … 433	0.960 [0.956, 0.963]	0.35
library Φ_ℓ, seconds (contended)	11 … 79	4.688 [4.633, 4.743]	−23.25

At $\ell \approx 2^{24}$: Vélu $1.01 \cdot 10^8$ Meq, √élu $2.45 \cdot 10^7$ (4.1× fewer); √élu passes Vélu near $\ell \approx 1000$. √élu's 0.82–0.86 instead of 0.5 is Karatsuba inside the trees ($\sqrt{\ell}^{1.585}$). Generic finding $2.6 = k \log p$ ladder steps × Karatsuba $\mathbb{F}_{p^k}$ products.



Points above the ℓ_max bar: no conductor gap that large exists, so no crossover.

Figure 2: Derived crossover ℓ^* — EXTRAPOLATED from the 61-bit counts (6 M per rho step, $p \approx r$, ladder phases scaled by $\log_2 r/61$). Bars: $\ell_{\max} = 2\sqrt{p}$, the largest prime a conductor gap can contain.

Derived crossover table (extrapolation, g = 6 M per rho step)

model	$r = 2^{64}$	$r = 2^{128}$	$r = 2^{256}$
rho (F_p mults)	$2^{34.4}$	$2^{66.4}$	$2^{130.4}$
$\ell_{\max} = 2\sqrt{p}$	2^{33}	2^{65}	2^{129}
rational, Vélu	$2^{31.9}$	$2^{63.9}$	$2^{128.0}$
rational, √élu (fit)	$2^{35.9} > \ell_{\max}$	$2^{74.8} > \ell_{\max}$	$2^{152.7} > \ell_{\max}$
rational, √élu ($\sqrt{\ell}\log^2 \ell$ model)	$2^{40.7} > \ell_{\max}$	$2^{99.5} > \ell_{\max}$	$2^{222.9} > \ell_{\max}$
generic, fit	$\mathbf{2^{9.6}}$	$\mathbf{2^{21.5}}$	$\mathbf{2^{45.7}}$
generic, $\ell^2 \log \ell$ model	$2^{9.8}$	$2^{24.7}$	$2^{55.6}$

Kernel over a small extension $\mathbb{F}_{p^k}$ ($r = 2^{128}$, derived): $\ell^* = 2^{99.5}, 2^{95.1}, 2^{91.5}, 2^{87.9}, 2^{84.3}, 2^{81.4}, 2^{78.5}$ for $k = 1, 2, 4, \ldots, 64$ — all above $\ell_{\max} = 2^{65}$.

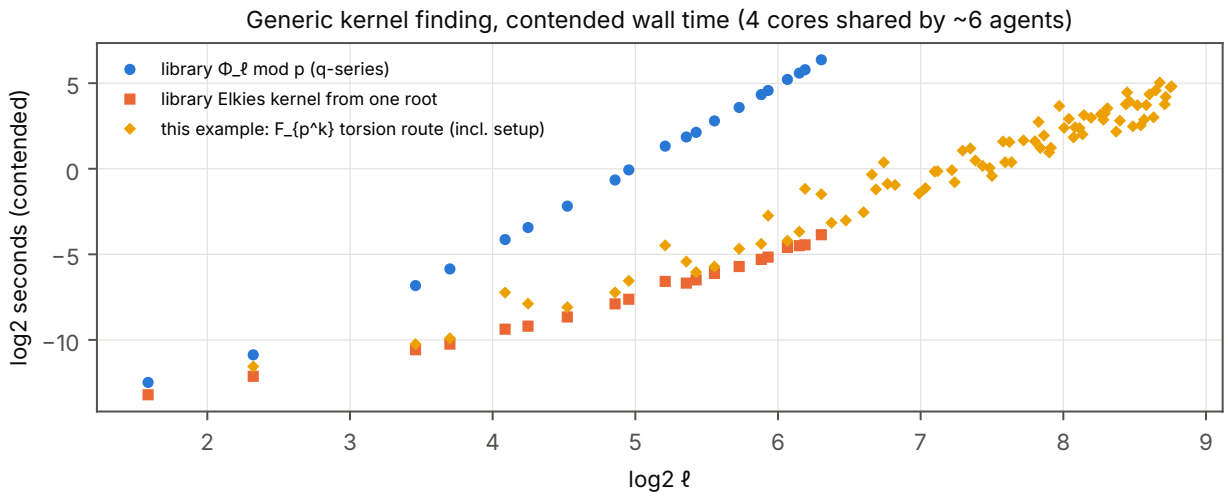


Figure 3: Generic kernel finding, contended wall time: library Φ_ℓ route ($\ell \le 79$) and the $\mathbb{F}_{p^k}$ torsion route of this example ($\ell \le 433$).

Verdict

Confirmed, sharpened. With a generic kernel ($k \approx \ell/2$) the transfer costs more than rho from $\ell \approx 2^{10}$ ($r = 2^{64}$), 2^{22}–2^{25} ($r = 2^{128}$), 2^{46}–2^{56} ($r = 2^{256}$); the cost is driven by the kernel's field degree k, not by ℓ itself. With a kernel point over $\mathbb{F}_p$ or a small extension there is **no** margin: with $\sqrt{}$élu no $\ell \le 2\sqrt{p}$ reaches rho, and Vélu reaches it only at $\ell \approx \sqrt{r}$. Prop. 3.3's "$\ell \gtrsim 2^{60}$ for a 128-bit group" is loose for the generic case and wrong for the rational one. For a cofactor-1 target the rational case needs $\ell \mid \#E$ or $\#E^{\text{twist}}$ over a small extension, which is unusual.

Limits

61-bit toy field; larger r is extrapolation (generic: measured $\ell \le 433$). Karatsuba, no FFT: the $\tilde{O}$ rows are calibrated models. Not implemented: ψ_ℓ factoring, Sutherland's $\Phi_\ell(j, Y)$ evaluation. Only descending edges from the crater; one discriminant (-7); instances restricted to $v_\ell = 2$. Additions and memory not counted. Conductor `check` was not available in this environment.